

Q1.What is Data Loading in Power BI and why is it considered the first step of analysis?

ANS : **Data Loading in Power BI** is the process of importing or connecting data from different sources into Power BI so it can be used for reporting, visualization, and analysis.

After loading, the data becomes available inside Power BI for cleaning, transformation, modeling, and visualization.

Why is it considered the first step of analysis?

Data loading is considered the first step because analysis cannot begin until the data is available inside the system. It provides the foundation for all later stages of business intelligence.

Reasons:

- 1) Brings data into Power BI
- 2) Combines data from multiple sources
- 3) Prepare data for transformation
- 4) Enables reporting and visualization
- 5) Supports accurate decision-making

Q2.Explain the difference between “Load” and “Transform Data” in Power BI.

ANS:

Feature	Load	Transform Data
Meaning	Imports data directly into Power BI.	Opens Power Query Editor to clean and modify data before loading.
Purpose	Quickly bring data into the model for analysis.	Prepare and improve data quality for accurate analysis.
Data Changes	No major changes are made to the data.	Data can be filtered, cleaned, merged, split, renamed, etc.
Tool Used	Loads data into Power BI Data Model.	Uses Power Query Editor .

Processing Time	Faster because data is loaded immediately.	Takes extra time due to transformation steps.
Use Case	When data is already clean and ready for reporting.	When data contains errors, duplicates, missing values, or inconsistent formats.
Output	Data becomes available directly for reports and visuals.	Transformed data is loaded after modifications are applied.

Q3. What is a Fact Table and a Dimension Table? Give examples from the dataset.

ANS :

Fact Table : A **Fact Table** is the main table in a data warehouse or Power BI model that stores **measurable business data** such as sales, quantity, profit, revenue, or transactions.

Example:

OrderID	ProductID	CustomerID	DateID	Quantity	SalesAmount
101	P01	C01	D01	2	5000
102	P02	C02	D02	1	3000

Dimension Table : A Dimension Table stores descriptive information about business entities such as customers, products, employees, or dates.

Example : **Product Dimension**

ProductID	ProductName	Category
P01	Laptop	Electronics

P02

Mobile

Electronic

Q4. Why is Star Schema preferred over Snowflake Schema in Power BI?

ANS : STAR SCHEMA: In a Star Schema:

- One central **Fact Table** is connected directly to multiple **Dimension Tables**.
- Dimension tables are not further divided.

SNOWFLAKE SCHEMA: In a Snowflake Schema:

- Dimension tables are further normalized into multiple related tables.

Reasons Why Star Schema is Preferred in Power BI

Reason	Explanation
Better Performance	Fewer joins are required, so Power BI queries run faster.
Simpler Design	Easy to understand and manage for developers and analysts.
Faster Report Creation	Relationships are straightforward, making visualization building easier.
Improved DAX Calculations	Measures and calculations work more efficiently with simpler relationships.
User-Friendly	Business users can easily understand dimensions and facts.

Optimized for BI Tools

Power BI's VertiPaq engine performs best with denormalized star structures.

Q5. Identify and remove duplicate records based on Date, Country, and State.